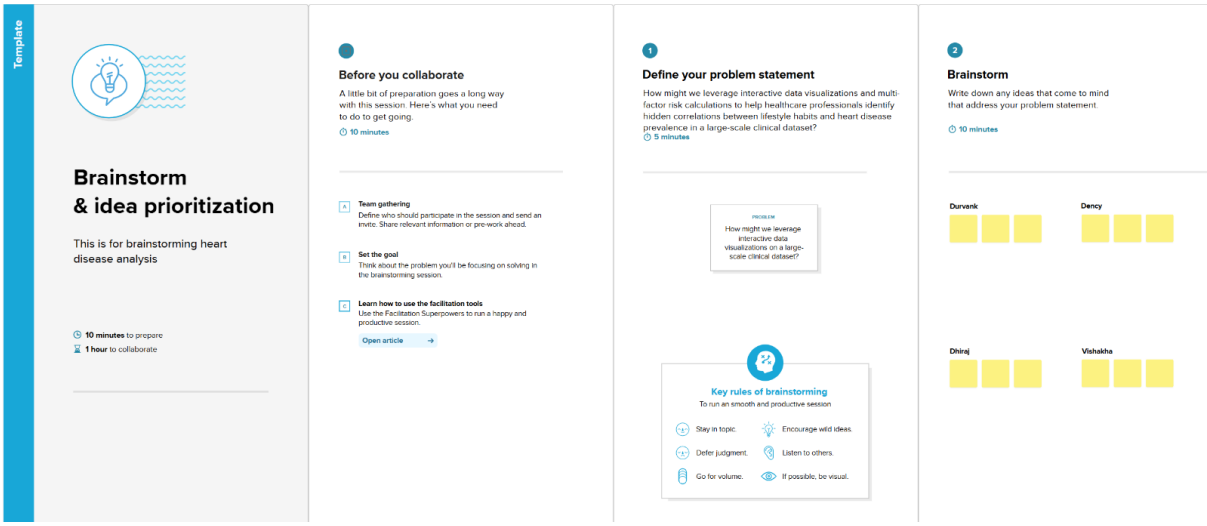
IDEATION PHASE

Brainstorm & Idea Prioritization

Date	02 March 2026
Team ID	
Project Name	Heart Disease Analysis
Maximum Marks	4 Marks

Step-1: Team Gathering, Collaboration and Select the Problem Statement

How Might We (HMW) Statement: How might we interactive data visualizations and multi-factor risk calculations to help healthcare professionals identify hidden correlations between lifestyle habits and heart disease prevalence in a large-scale clinical dataset?

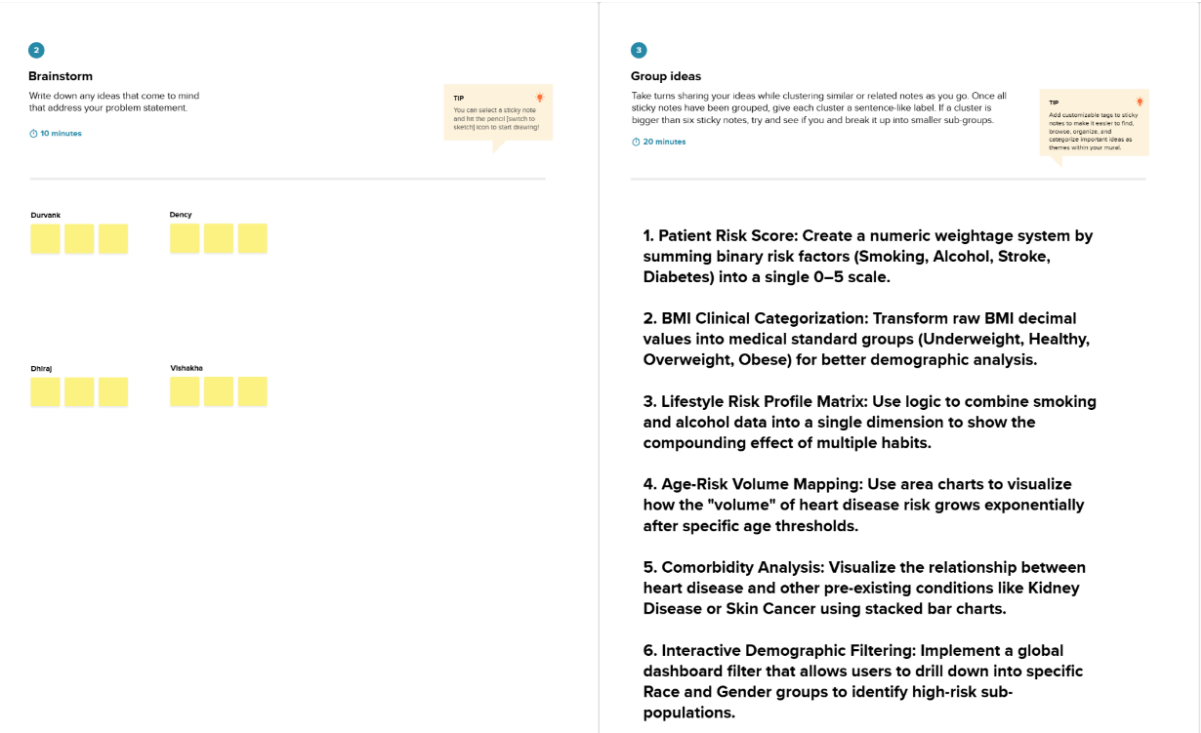


Step-2: Brainstorm, Idea Listing and Grouping

Idea Listing:

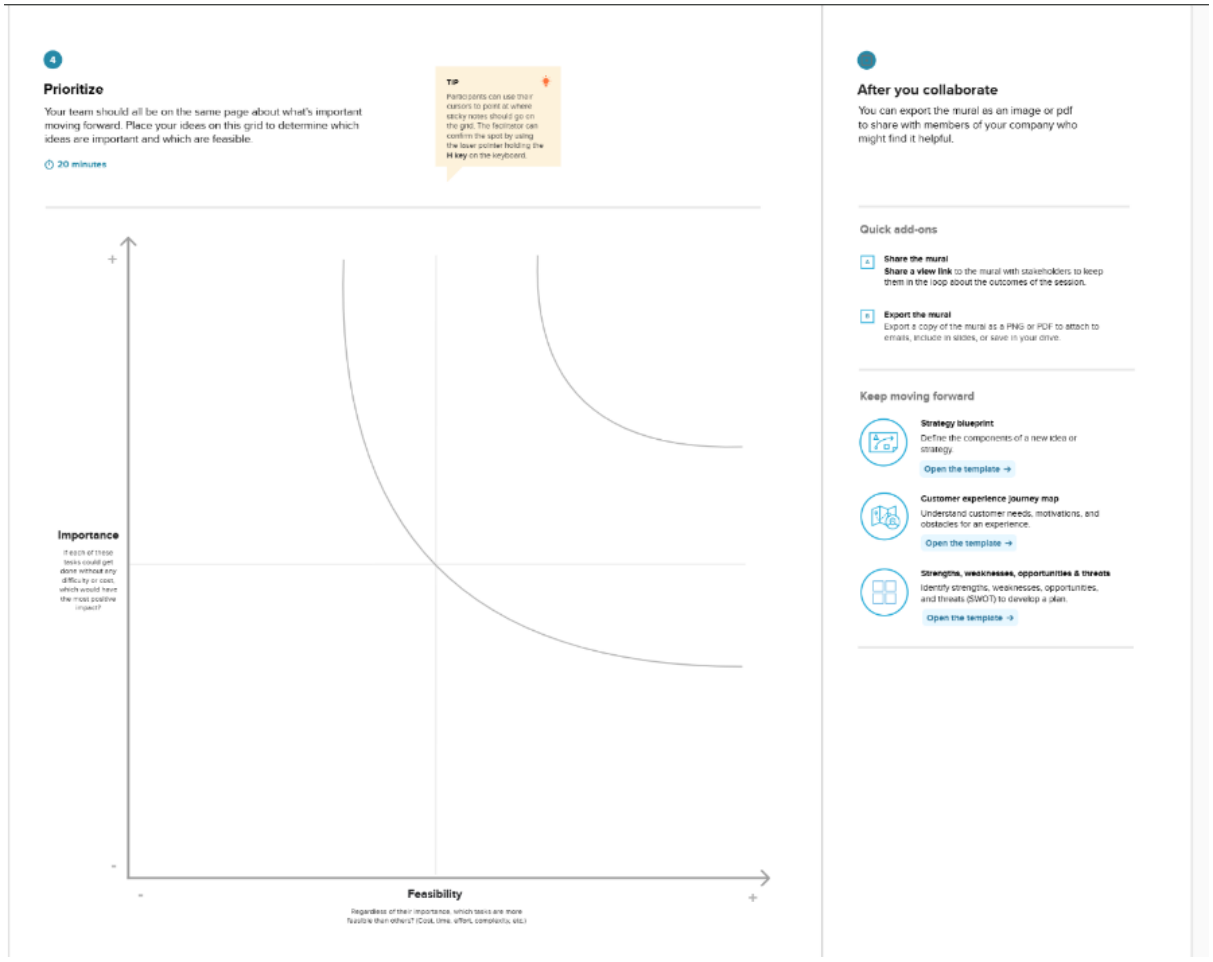
1. Patient Risk Score: Create a numeric weightage system by summing binary risk factors (Smoking, Alcohol, Stroke, Diabetes) into a single 0–5 scale.
2. BMI Clinical Categorization: Transform raw BMI decimal values into medical standard groups (Underweight, Healthy, Overweight, Obese) for better demographic analysis.

3. Lifestyle Risk Profile Matrix: Use logic to combine smoking and alcohol data into a single dimension to show the compounding effect of multiple habits.
4. Age-Risk Volume Mapping: Use area charts to visualize how the "volume" of heart disease risk grows exponentially after specific age thresholds.
5. Comorbidity Analysis: Visualize the relationship between heart disease and other pre-existing conditions like Kidney Disease or Skin Cancer using stacked bar charts.
6. Interactive Demographic Filtering: Implement a global dashboard filter that allows users to drill down into specific Race and Gender groups to identify high-risk sub-populations.



Step-3: Idea Prioritization

1. High Importance / High Feasibility (Quick Wins):
   - Patient Risk Score Calculation: High value for clinicians and easy to implement via Tableau calculated fields.
   - BMI Standard Categorization: Essential for clinical accuracy and simple to execute in SQL or Tableau.

- Age and Gender Trend Analysis: The most common requests from stakeholders and easy to visualize.

2. High Importance / Low Feasibility (Major Projects):
    - Real-time Database Integration: Highly important for live clinical monitoring but requires significant backend infrastructure and API management.
3. Low Importance / High Feasibility:
    - Basic Record Counting: Easy to do but provides limited "actionable insight" compared to calculated risk rates.
4. Low Importance / Low Feasibility:
    - Manual Data Entry Forms: High effort to build and maintain compared to using the existing automated database connection.